Quinton Wilson

Data Scientist, Healthcare Analytics | wilsonq0629@gmail.com | linkedin.com/in/quintonmwilson

Summary

Data scientist with 8 years of experience building predictive models and validated data pipelines across regulated government, distribution and animal health data. Currently building a 30-day hospital readmission risk model on MIMIC-IV EHR data (encounters, vitals, labs, ICD-10 codes) with SHAP explainability. Built daily population forecasts (R^2 0.94) supporting emergency services staffing for Surf City, NC, and anomaly detection on 100M+ records in a regulated environment. M.S. in Applied Data Science (UNC Chapel Hill, 2025); CITI human subjects and HIPAA privacy trained.

Core Skills

- **Machine Learning & Statistics:** Python (Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, PyTorch), predictive modeling, classification, regression, time series forecasting, imbalanced-data modeling (SMOTE), explainable ML (SHAP), hypothesis testing
- **Healthcare Data:** EHR analytics (MIMIC-IV), cohort and outcome definition, ICD-10 diagnosis codes, longitudinal patient data, HIPAA privacy (CITI)
- **SQL & Data Engineering:** Advanced SQL, GCP/BigQuery, Snowflake, ETL, data warehousing, data integrity validation, Azure (AZ-900)
- **Visualization & Reporting:** Tableau, Power BI, Alteryx (Certified), stakeholder dashboards
- **NLP & Deep Learning:** Graduate coursework with Hugging Face Transformers and GPT models

Healthcare Analytics Project

Hospital Readmission Prediction (MIMIC-IV) | In progress

- Defining an adult inpatient cohort and 30-day readmission outcome from longitudinal EHR data (encounters, vitals, labs, ICD-10 diagnosis codes), with the extraction and feature pipeline in SQL on Google BigQuery and Python.
- Building readmission risk models (XGBoost, Random Forest) with SHAP explainability to surface interpretable risk drivers for clinical and operational stakeholders.

Professional Experience

Data Scientist, Covetrus (animal health technology) | 02/2026 – Present | Remote

- Designed and ran A/B and multivariate tests on offers and creative across customer segments, including lapsed and at-risk customers, measuring impact on auto-ship enrollment, gross profit and customer allocation cost.
- Built a regression forecasting and quadratic optimization model for monthly and sale-event marketing spend and 4-segment audience allocation; cut planning time ~30% with forecast spend within ~5% of allocated budget.
- Segmented ~15M consumers per send for sitewide and flash-sale campaigns, supporting ~$4.5M in monthly gross profit.
- Refactored legacy SQL segmentation logic (~25% faster runtime) and benchmarked it against the next-gen model, delivering a migration recommendation to leadership.

Data Analyst I, Southern Glazer's Wine & Spirits | 11/2021 – 02/2026 | Miramar, FL

- Built Python demand forecasting models (~12% MAPE), optimizing inventory allocation across 12 distribution centers.
- Applied SQL-based data mining and pattern analysis to 500K+ customer records, delivering recommendations that increased strategic project success rates by 20%.

- Engineered automated Tableau and Power BI dashboards, cutting executive reporting time by 35 hours/month for 15+ stakeholders.
- Partnered with sales and operations stakeholders to translate technical findings into business strategy, driving $2.3M in cost savings.

Senior Operations Analyst, Florida Department of Revenue | 08/2018 – 08/2021 | Tallahassee, FL

- Applied statistical anomaly detection to 100M+ regulated taxpayer records using SQL and Python, identifying $4.2M in revenue recovery opportunities.
- Built automated data-validation workflows across 50+ monthly compliance reports, improving dataset reliability 15% and reducing critical errors 8%.
- Built regression-based forecasting models projecting quarterly tax revenue within 3% accuracy, supporting $850M annual budget planning.
- Authored analytical frameworks and process documentation, standardizing reporting workflows for an 8-person analytics team.

Additional Projects

Visitor Forecasting for Surf City, NC | UNC capstone with Blue Nalu Analytics

- Machine learning developer on a 4-person team that replaced the city's SARIMA model with daily visitor forecasts, helping emergency services and city departments staff ahead of tourism surges.
- XGBoost model explained 94% of variance in daily out-of-market visitors (R^2 0.936, 10.5% MAPE, RMSE ~3,300 visitors), benchmarked against Random Forest and a transparent weighted year-over-year growth baseline.
- Engineered 25+ features from 5 years of daily foot traffic, event, weather and market data, selecting lags via ACF/PACF/AIC analysis and excluding variables unavailable at prediction time to prevent leakage.

Financial Fraud Detection | github.com/quintonmwilson/financial-fraud-detection

- With two classmates, compared neural network, XGBoost and Random Forest classifiers on highly imbalanced transaction data, testing SMOTE, undersampling, cost-sensitive learning and threshold tuning; XGBoost delivered the best fraud-class F1 (0.87, ROC-AUC 0.998) while training ~100x faster than the neural network (4s vs. 415s).

Education

- M.S., Applied Data Science, University of North Carolina, Chapel Hill | 11/2025
- B.S., Business Administration, Florida A&M University | 12/2017

Certifications

CITI Program, Human Subjects Research incl. HIPAA Privacy Protections (12/2025) | Alteryx Designer Core Certification (11/2023) | Microsoft Azure Fundamentals, AZ-900 (06/2023)